Limitations and Planned Evaluation: Delivery Dependencies, Kernel Scope, and Learner Outcomes

Delivery dependencies, fixed-kernel scope, and learner-outcome boundaries

Delivery substrate and kernel scope I

The local learner dashboard can be verified in artifact-only mode without Docker; live progress, ranking, and quiz completion still require a SkillTree service. When that service is available, the dashboard mounts the official learner display and separates local lesson material from official subject and quiz handoffs, but it does not provide a second source of learner truth. The REST contract is verified against a `pytest-httpserver` mirror of SkillTree's endpoints, and the live seed is documented and scripted.

Delivery substrate and kernel scope I

The implemented kernels cover discrete pedagogical cases, reference variational examples, and bounded predictive-coding or planning demonstrations; they do not provide a continuous, scalable, robust, or deep predictive-coding inference engine. Reactive message passing, RxInfer, coupled free energy, active predictive coding, divide-and-conquer predictive coding, generalized filtering, generalized free energy, hierarchical multi-step planning, and richer reinforcement-learning comparisons remain cited roadmap context rather than hidden implementation claims (Bagaev and Vries 2023; Bagaev et al. 2023; Nelson et al. 2025; Rao et al. 2024; Sennesh et al. 2024; Friston et al. 2008; Friston et al. 2010; Parr and Friston 2019). The RL comparison matters because probabilistic-inference views of control overlap with, but are not identical to, active-inference policy selection (Levine 2018). Psychology-and-psychiatry Active Inference

reviews likewise support cautious application language rather than theory-level validation claims (Badcock and Davey 2024). Finally, the platform's contribution is the verifiability and provenance of curriculum content, not a measured claim about learning outcomes: it does not evaluate teaching effectiveness, retention, or transfer, and no efficacy study is reported here. Microlearning studies, epistemic-curiosity evidence, learning-analytics reviews, and learning-analytics intervention meta-analyses motivate the proposed study design but do not substitute for data from this curriculum, and learning analytics also introduces privacy and data-protection obligations before progress exports can be treated as evidence (Monib et al. 2025; Rof et al. 2024; Emin et al. 2026; Liu et al. 2025; Hernández-Campos et al. 2025; Liu and Khalil 2023). Whether this provenance discipline actually improves how people learn the framework

remains future work, outside the scope of what the code, citations, and validators establish. The preregistered pilot design in `experiment_plan.yaml` is therefore a setup artifact: it defines consent, privacy, SkillTree progress-export, pre/post, delayed-retention, and analysis boundaries, and follows design-based research framing for studying a curriculum artifact in context, but it is not learner-outcome evidence until real data are collected and added to the claim-support record (Design-Based Research Collective 2003; Barab and Squire 2004).